Federated Learning for Privacy-Preserving Video Anomaly Detection Across Two Camera Sites

Nishant Ajit, Rohith Elindram, Ashwin Kumar, Pushpak
SCOPE, Vellore Institute of Technology, Vellore, 632014, India

Abstract

Cities such as Singapore run large camera networks, and sending all of that footage to one central server raises privacy concerns. Federated learning avoids this, because each camera site trains on its own video and shares only model weights. We tested how well that holds up for video anomaly detection. UCSD Ped1 and Ped2, which are public pedestrian datasets were used for two camera sites. Each site trained a small convolutional autoencoder on normal footage, a server combined the two models with federated averaging (FedAvg) over 10 rounds, and frames were scored by reconstruction error. We compared this against the same model trained on pooled data (centralised training) and against models trained at each site alone, using the frame-level area under the ROC curve (AUC) over five runs. The federated model reached a mean AUC of 0.731, close to the 0.737 of centralised training and clearly above the single-site models. Training exchanged 52.6 MB of model weights in total.

Keywords: federated learning; anomaly detection; surveillance video; autoencoder; data privacy; UCSD pedestrian dataset

1. Introduction

Video cameras today can be seen in numerous locations including streets, subway stations, and schools. Most of the footage recorded is uninteresting, thus creating the need for a system that identifies unusual events, such as a biker using a sidewalk or a car parked in a pedestrian zone. Video anomaly detection is based on the idea that only normal footage should be used to train the model and everything that is mistakenly classified as normal should be flagged as uncommon.

Singapore illustrates the value of privacy in this context. Its Smart Nation initiative has introduced extensive video surveillance and established a working framework of the Personal Data Protection Act that makes it clear what can and cannot be done with personal data. However, a plan to test facial recognition technology on lamp posts in 2018 raised public fears about privacy, which demonstrates that even cities with the necessary source of information are not ready to adopt a surveillance system that sends all recorded footage to one main server.

In federated learning, the original video footage is retained at the location of collection and only the trained model weights are shared with the server, which averages them in order to create a single model over many rounds. This raises the question of what the trade-off is, in terms of detection accuracy, by having data at different sites rather than in a single pool for training. We decided to quantify this trade-off under controlled conditions whereby the study could be repeated from start to finish and see whether there are any advantages of using federated training compared to using the footage collected by the cameras on their own. To do this, we used two sites implementing the FedAvg method whereby the two cameras that were used, UCSD Ped1 and Ped2, were used separately without exchanging the video footage. The results of the federated training were compared to those of the centralized training and single site training with the same model and number of training epochs in five runs. The results obtained with federated training were almost as good as those obtained with centralized training.

The federated model came within 0.006 mean AUC of centralised training and beat single-site training.

2. Literature Review

2.1 Video anomaly detection

Chandola, Banerjee and Kumar survey anomaly detection across many fields. The UCSD pedestrian dataset, released by Mahadevan and colleagues in 2010, is one of the standard benchmarks for video anomaly detection. Hasan and colleagues later showed that a convolutional autoencoder trained only on normal frames gives high reconstruction error on unusual ones, which made reconstruction error usable as an anomaly score. Accuracy has since been pushed up in two main ways, either by predicting the next frame instead of rebuilding the current one, as Liu and colleagues did, or by adding a memory of normal patterns, as in MemAE by Gong and colleagues. Sultani and colleagues took weak, video-level labels to large real-world datasets. Wu and colleagues review all of these families in a recent survey. Every one of them trains on video gathered in one place.

2.2 Federated learning

FedAvg, introduced by McMahan and colleagues, is the standard federated learning method. A few epochs of local training is performed by each client, and the server takes a weighted average of the returned weights. A large survey led by Kairouz lays out the field's open problems. Its weak spot is non-IID data. Zhao and colleagues showed that accuracy can drop when the data at different clients look different, and separate camera scenes are an obvious case of this. FedProx from Li's team and SCAFFOLD from Karimireddy's were built to reduce this drop. Shared weights are not fully private either. Zhu, Liu and Han recovered training images from shared gradients.

2.3 Federated video anomaly detection

Federated learning has reached video anomaly detection only recently. Singh and colleagues applied it to autoencoder-based video anomaly detection. CLAP, by Al-lahham and colleagues, trains an unsupervised detector across owners without moving their video. FedVAD, by Qi and colleagues, groups similar clients and uses a large language model to guide the global model. Wang and colleagues paired federated learning with weak labels and text prompts for a CLIP-based detector, while Su and colleagues used a mixture of local and global experts to cope with differences between camera scenes. Liu and colleagues survey these and other privacy-preserving approaches. Together these methods show that federation can work for video. The more recent ones, though, rely on large pre-trained models or extra labels.

2.4 Research gap

Table 1 summarises representative studies. Most video anomaly detection work assumes centralised data, and the recent federated methods are large and complex. Few studies report, for a lightweight model, the accuracy gap between federated and centralised training of the same network together with a comparison against training at one site alone.

Study	Approach	Dataset / setting	Limitation
Hasan et al., 2016	Convolutional autoencoder, reconstruction error	UCSD Ped1, Ped2, Avenue, Subway	Assumes all video is pooled centrally
Liu et al., 2018	Future frame prediction	UCSD Ped1, Ped2, Avenue, ShanghaiTech	Centralised; larger model
Gong et al., 2019	Autoencoder with memory	UCSD Ped2, Avenue,	Centralised training only

Study	Approach	Dataset / setting	Limitation
	module (MemAE)	ShanghaiTech	
Al-lahham et al., 2024	Collaborative unsupervised detection (CLAP)	Large surveillance video benchmarks	Complex pipeline; no single-site comparison
Qi et al., 2024	Federated detection guided by a language model (FedVAD)	Surveillance benchmarks split across clients	Depends on large pre-trained models
Wang et al., 2025	Federated weak supervision with text prompts	Weakly labelled surveillance benchmarks	Needs video-level labels and CLIP
Su et al., 2025	Mixture of local-to-global experts	Scenario-specific clients	Needs video-level labels; complex design
This work	FedAvg over a small convolutional autoencoder	UCSD Ped1 and Ped2 as two sites	Addressed: same-budget comparison with centralised and single-site training

Table 1. Comparison of related work and the gap addressed by this study.

3. Methodology

3.1 Problem formulation

We have $K = 2$ camera sites, each holding unshareable footage. The autoencoder maps a frame x to a reconstruction $\hat{x}$ and is trained with the mean squared error over all $H \times W$ pixels in Eq. (1).

$$\ell(x) = (1 / HW) \, \|x - \hat{x}\|^2 \tag{1}$$

At test time, the error of Eq. (1) on a frame is its anomaly score.

3.2 Data

We used the public UCSD Anomaly Detection Dataset, containing two outdoor pedestrian scenes, each filmed by a fixed camera. Ped1 has 34 training clips and 36 test clips of 200 frames each, at 238 × 158 pixels. Ped2 has 16 training clips and 12 test clips of 120 to 180 frames, at 360 × 240 pixels. The training clips show only normal walking. In the test clips cyclists, skaters and small carts also appear, and each test frame is labelled normal or abnormal.

Ped1 acted as Site 1 and Ped2 as Site 2. Video data between site 1 and site 2 is not exchanged at any point in time. Site 1 had 6,800 training frames and 7,200 test frames, 4,045 (56.2%) of them abnormal. Site 2 had 2,550 training frames and 2,010 test frames, 1,648 (82.0%) of them abnormal. The two sites differ in camera angle, image size and amount of data. Every frame was converted to grayscale, resized to 64 × 64 pixels and scaled to [0, 1]. 64 x 64 was chosen to provide a common input to the shared model as both sites record at different resolutions. Small frames also keep the network lightweight. Labels were used only for evaluation. One test frame (Ped1, clip Test017, frame 142) is corrupt, so we replaced it with the previous frame.

3.3 Proposed approach

Figure 1 shows the full system. The encoder has three convolution layers with 32, 64 and 128 filters, each halving the image size, so a 64 × 64 frame ends up as a 128 × 8 × 8 code. A mirrored decoder of three transposed convolution layers follows and finishes with a sigmoid, which keeps its output in [0, 1] like the input. In total the model has 329,025 parameters. We left out batch normalisation so that every weight could be averaged directly.

Training followed FedAvg. In each round the server sent the current global weights to both sites. Each site trained them for 3 epochs on its own frames and sent the new weights back, and the server averaged the two sets with Eq. (2), giving each site a share equal to its number of training frames. We ran 10 such rounds. Three local epochs per round put the federated budget at 30 epochs per site, the same as the other two training methods.

$$\theta_{\text{global}} = (n_1\,\theta_1 + n_2\,\theta_2) / (n_1 + n_2) \tag{2}$$

Here θ_1 and θ_2 are the weights returned by the two sites, and n_1 and n_2 are their numbers of training frames. For testing, each frame was scored with the error of Eq. (1). As is common on this dataset, the scores were then rescaled to [0, 1] within each test clip using Eq. (3), where e_t is the error of frame t and the minimum and maximum are taken over the clip. The labels play no part in this step.

$$s_t = (e_t - e_{\min}) / (e_{\max} - e_{\min}) \tag{3}$$

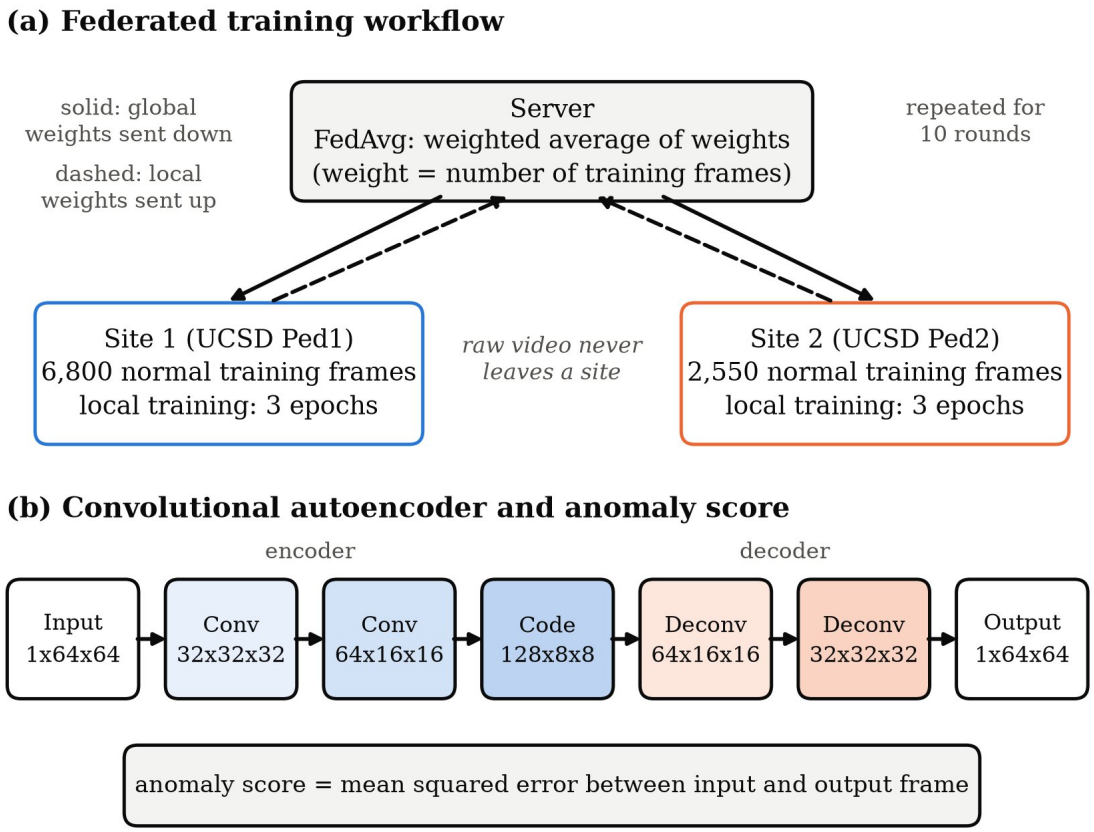


Figure 1. Overview of the system. (a) In each of 10 rounds the server sends the global autoencoder weights to Site 1 (UCSD Ped1) and Site 2 (UCSD Ped2). Each site trains for 3 epochs on its own normal frames, and the server averages the returned weights in proportion to each site's number of training frames Only weights are exchanged. (b) The convolutional autoencoder (Conv denotes convolution and Deconv transposed convolution) turns a 64 × 64 grayscale frame into a 128 × 8 × 8 code and back. A frame's anomaly score is its mean squared reconstruction error.

3.4 Experimental setup

We trained the same model in three ways, each with 30 passes over each site's data. Federated training ran FedAvg for 10 rounds with 3 epochs per site per round. Centralised training pooled the 9,350 training frames of both sites and trained one model for 30 epochs. It ignores privacy and serves as the reference. In single-site training, each site trained its own model for 30 epochs on its own frames only.

All models used the Adam optimiser of Kingma and Ba with a learning rate of 0.001, a batch size of 64 and the loss of Eq. (1). Our main metric is the frame-level area under the ROC curve (AUC) described by Fawcett, computed separately for each site over all of its test frames, and we also report the mean of the two site AUCs. Communication cost follows Eq. (4), where P is the number of parameters at 4 bytes each, the factor 2 counts sending and receiving, K is the number of sites and R the number of rounds.

$$C = P \times 4 \text{ bytes} \times 2 \times K \times R \tag{4}$$

Each experiment was repeated 5 times with random seeds 42 to 46, and we report the mean ± standard deviation. Runs with the same seed give exactly the same result. To check whether a difference between two methods is larger than run-to-run noise, we used a paired t-test on the mean AUC. The code was written in Python 3.11 with PyTorch 2.14 and scikit-learn 1.9. All runs used one NVIDIA GeForce GTX 1650 Ti laptop GPU, with both sites simulated on the same machine. A federated run took about 63 s and a centralised run about 57 s.

4. Results

4.1 Detection accuracy

Table 2 gives the results. The federated model reached a mean AUC of 0.731 ± 0.011, against 0.737 ± 0.006 for centralised training. That gap of 0.006 is smaller than the run-to-run variation, and the t-test did not find it significant ($p = 0.28$). Per site the picture splits, with the federated model doing better on Site 1 (0.709 against 0.694) and centralised training doing better on Site 2 (0.779 against 0.753).

Single-site training was clearly weaker. The Site 1 model reached 0.696 on its own site but only 0.515 on Site 2, close to chance. The Site 2 model held up better, with 0.718 on its own site and 0.668 on Site 1. Against both single-site models the federated model had the higher mean AUC ($p < 0.001$ and $p = 0.007$), and in 4 or more of the 5 runs it also beat each single-site model on that model's own site.

Method	Site 1 AUC (Ped1)	Site 2 AUC (Ped2)	Mean AUC	p-value (vs. federated)
Single site, Site 1 model	0.696 ± 0.011	0.515 ± 0.061	0.605 ± 0.028	< 0.001
Single site, Site 2 model	0.668 ± 0.005	0.718 ± 0.016	0.693 ± 0.008	0.007
Centralised (pooled data)	0.694 ± 0.004	**0.779 ± 0.013**	**0.737 ± 0.006**	0.283
Federated (this work)	**0.709 ± 0.006**	0.753 ± 0.019	0.731 ± 0.011	Reference

Table 2. Frame-level ROC-AUC of each training method on the test clips of both sites. Values are mean ± standard deviation over 5 runs. Single-site models are trained on one site and tested on both. p-values come from a paired t-test on the mean AUC against the federated model. Best result in each column in bold.

4.2 Training rounds and communication cost

Figure 2 tracks the mean AUC of the federated model round by round. It started at 0.668 after round 1, rose to 0.736 by round 7 and then levelled off, ending at 0.731. The spread across runs also narrowed as training went on. Each round exchanged 5.26 MB, which comes to 52.6 MB over the full 10 rounds.

Stopping at round 7 would have needed 36.9 MB. For comparison, the raw training video of both sites is about 476 MB.

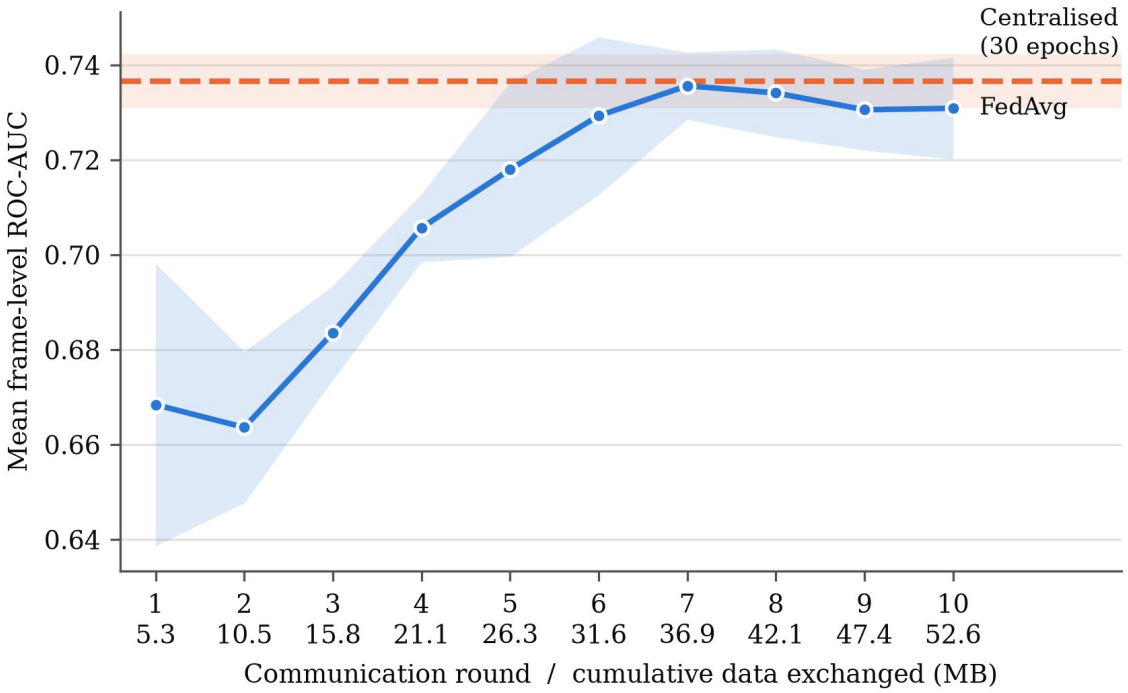


Figure 2. Mean frame-level ROC-AUC (average of Site 1 and Site 2) of the federated model after each round. The solid line is the mean over 5 runs and the shaded band shows ± one standard deviation. The dashed line shows the centralised model after 30 epochs. The second row of x-axis labels gives the total data exchanged so far, in megabytes (MB).

4.3 Example detections

Figure 3 comes from the saved seed-42 federated model, so it shows a single run. Among Ped2 test clips where normal and abnormal frames each make up at least a quarter of the frames, the clip with median per-clip AUC was taken, which was clip 002.

The normal frame shown is the middle normal frame of that clip (frame 48). For the abnormal one we deliberately took the highest-scoring abnormal frame (frame 159), the clearest case the clip offers. Even so, the gap is small, with a mean squared error of 8.7×10^{-4} against 7.6×10^{-4}. In both error maps the error sits on the people in the scene, whom the decoder blurs regardless of what they are doing. At 64 × 64 an abnormal object covers only a handful of pixels, and that may be part of why the frame-level AUCs in Table 2 are modest.

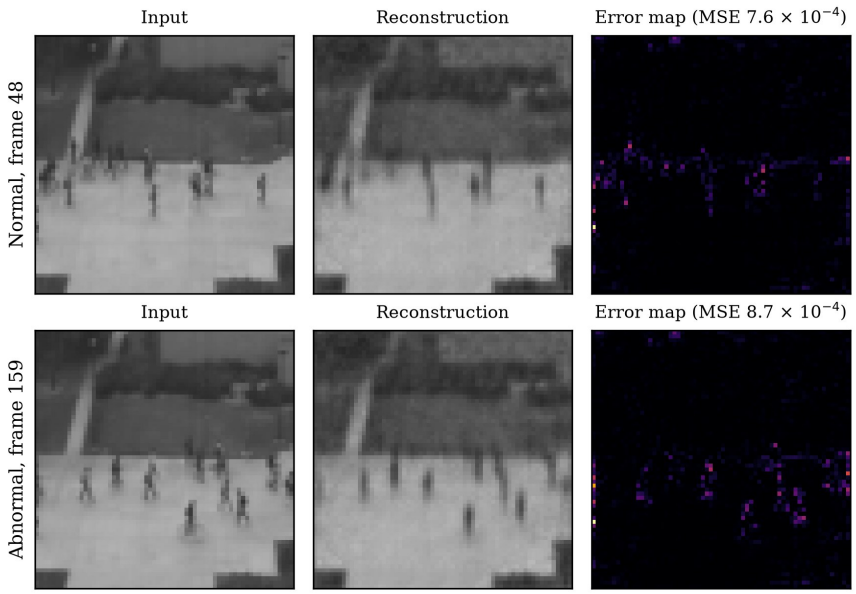


Figure 3. Input, reconstruction and squared-error map for two frames of Ped2 test clip 002, from the seed-42 federated model. The top row is the middle normal frame and the bottom row the highest-scoring abnormal frame. Both error maps use the same colour scale.

5. Discussion

Federated training came close to centralised training. The mean AUC gap of 0.006 falls inside the run-to-run variation, so on this task keeping the video at each site seems to have cost little accuracy, although with five runs and two sites we would not read much into the exact size of the gap. One likely reason is that both scenes show much the same content, namely a fixed camera, a walkway and people walking. A small autoencoder probably spends most of its capacity on these shared patterns, and averaging two models that have learned similar things should lose little. But the per-site results complicate this picture. Averaging shifted the balance between the sites, helping Site 1 and hurting Site 2.

The comparison with single-site training is where federation looks most useful. A model trained at one site did poorly at the other (the Site 1 model scored 0.515 on Site 2), and the federated model did better than either single-site model overall. For a city such as Singapore, this hints that camera sites with similar scenes could build one shared detector while keeping their footage on site.

Our AUC values are lower than published results on UCSD. Hasan and colleagues, for example, reported 0.810 on Ped1 and 0.900 on Ped2. Their model used larger frames and stacks of frames that capture motion, whereas ours looks at one small frame at a time, and we did not try to match their scores.

6. Conclusion

Within the limits of two sites and five runs, federated averaging let a small autoencoder learn a shared anomaly detector from UCSD Ped1 and Ped2 at little cost in accuracy, and the result did better than either site could manage alone. Next we want to test more sites with more varied scenes, add motion by feeding the model several frames at once, and add secure aggregation, as in the Bonawitz protocol, so that the server cannot see any single site's update.

Declarations

Data availability. The UCSD Anomaly Detection Dataset is publicly available from the Statistical Visual Computing Laboratory, UC San Diego (http://www.svcl.ucsd.edu/projects/anomaly/).

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